

CLEANING Silently With SOUND WAVES



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Often, we struggle to clean electronic gadgets and other gizmos, especially those having small, delicate, inaccessible parts. The proven solution to such a need is an ultrasonic cleaner. In order to understand how an ultrasonic cleaner works, we should look at the techniques behind it and its inner electronics.

Nothing but a bubble

At the heart of ultrasonic cleaning is a bubble. Bubbles are created by ultrasonic sound waves (having frequencies higher than audible sound) as these move through water. This is known as cavitation, which is simply the formation of bubbles (cavity) in water. Since bubbles created through cavitation are just empty spaces, there is nothing to keep these open. As a result, bubbles implode almost as fast as these are created.

In an ultrasonic cleaner, this happens millions of times per second. This constant implosion produces tremendous vacuum energy in the form of heat and pressure that gives enormous cleaning power to ultrasonics. When cavitation happens near a dirty object, vacuum action produced by those millions of bubbles constantly imploding creates a tiny pressure wave that reaches deep into every nook and cranny of even the most delicate items. This tiny pressure wave dislodges and breaks up the dirt and other contaminants and gently lifts it away. The result is very fast and effective cleaning.

The silent scrub

You can use ultrasonic cleaning for fast and perfect removal of contaminants from objects by immersing these in a tank of fluid flooded with ultrasonic waves. Ultrasonic energy enters the fluid within the tank and causes the rapid formation and implosion of minute bubbles. The bubbles rapidly increase in size until

these implode against the surface of the object immersed in the tank. The resulting enormous energy release lifts contamination off the surface and innermost recesses of even intricately-shaped parts.

Ultrasonic cleaning is the ultimate solution for cleaning contact lenses, jewellery and metal goods such as dental tools, watches and screws. With the right method, ultrasonic cleaning can thoroughly strip the object of dirt, dust, grime and other contaminants, both on the outer and inner surfaces, without causing any damage to the object.

The cleaning process is non-destructive and produces consistent cleaning results of the highest standard, time after time. Ultrasonic cleaning is an energy-efficient, environmentally-friendly and technically-feasible alternative to cleaning solvents that are ozone-depleting and have other undesirable characteristics. Reduced emissions, biodegradable waste and a safer working environment are among the many benefits of using an ultrasonic cleaner.

Cleaning machines

Nowadays, in addition to industry-grade equipment, lightweight (portable/desktop) ultrasonic cleaners are also available for day-to-day cleaning of objects. These cleaners come in a range of sizes that have different working frequencies. Desktop models with lower frequency (40kHz -70kHz) waves are good for home and dental equipment. Powerful cleaners capable of making ultrasonic frequencies as high as 200kHz (70kHz -200kHz) are expensive, but worth the investment for precious jewellery and objects with fragile parts, such as watches and contact lenses.

Basic components of an ultrasonic cleaner include an ultrasonic transducer, an ultrasonic generator and a tank filled with aqueous solution. The key component is the transducer that generates the high-frequency mechanical energy. Outer body of the ultrasonic cleaner consists of a stainless

Fig. 1: Portable ultrasonic cleaner

